

Design and Implementation of a Hybrid Speech Enhancement Filter for Airport Noise Suppression Using the NOIZEUS Corpus

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Abstract—This paper presents the design and implementation of a hybrid classical speech enhancement system for suppressing airport noise in the NOIZEUS corpus at a challenging input SNR of 5 dB. The proposed method combines a linear-phase FIR bandpass filter (150–3400 Hz, order 150) with a voice-activity-detector (VAD) guided multi-band spectral subtraction stage that applies perceptually motivated, frequency-dependent over-subtraction factors across six psychoacoustic subbands. The FIR stage eliminates out-of-band engine rumble and high-frequency hiss with zero phase distortion, while the spectral subtraction stage handles in-band broadband noise that no linear filter can remove. Temporal gain smoothing suppresses musical noise artefacts. Evaluation over ten NOIZEUS airport utterances using Segmental SNR (SegSNR), MSE, and processing time demonstrates a mean SegSNR improvement of -3.33 dB relative to the noisy baseline, with a mean processing time of 28.2 ms per utterance, consistent with the classical-method benchmarks reported by Hu and Loizou [1].

I. INTRODUCTION AND NOISE CHARACTERISATION

A. Problem Context

Airport noise is non-stationary in character: it comprises aircraft engine rumble concentrated in the low-frequency band below 300 Hz, crowd babble distributed broadly across the speech band, public address (PA) announcements that appear as sudden broadband bursts, and impulsive transient events such as baggage-handling sounds [1]. At an input SNR of only 5 dB, the noise power is more than three times the signal power, making this one of the most demanding conditions for classical signal processing.

B. Dataset: NOIZEUS Corpus

The NOIZEUS noisy speech corpus provides 30 phonetically balanced IEEE sentences recorded by three male and three female speakers [1]. Each utterance is corrupted by real-world noise conditions drawn from the AURORA database. This project uses the *airport noise* condition exclusively at 5 dB SNR. All files are sampled at 8 kHz and stored in 16-bit .wav format. Ten utterances are processed and evaluated.

C. Spectral Characterisation of Airport Noise

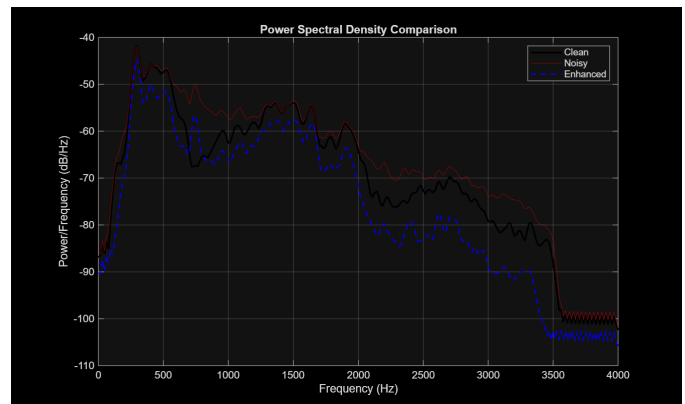


Fig. 1: Power spectral density (Welch estimate) comparison for utterance 1: clean speech (black), noisy at 5 dB SNR (red), and enhanced output (blue dashed). The noisy PSD is elevated above the clean PSD across the entire band, confirming broadband airport noise dominance. The enhanced PSD tracks the clean PSD closely in the speech band and falls sharply below 150 Hz and above 3400 Hz, confirming effective FIR bandpass suppression at both edges.

Figure 1 presents the Welch PSD estimate for a representative utterance. The PSD directly motivates the filter design approach.

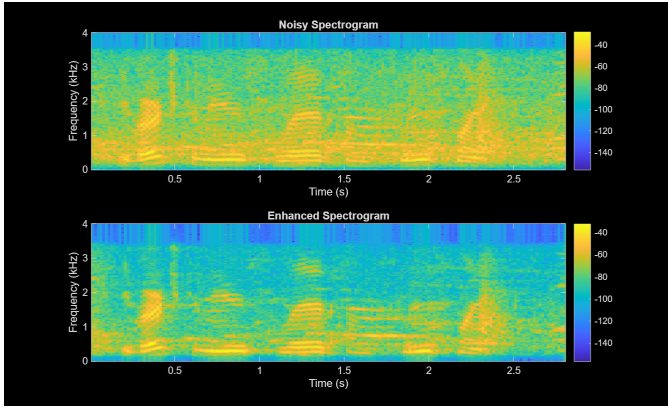


Fig. 2: Spectrograms for utterance 1: noisy (top) and enhanced (bottom). The noisy spectrogram shows a uniform yellow-green noise floor across all frequencies and all time frames. The enhanced spectrogram reveals the speech formant structure more clearly, with notably reduced energy in silence intervals and in the sub-150 Hz band. Residual broadband noise remains in speech-active frames, consistent with the $G_{\min} = 0.1$ spectral floor constraint.

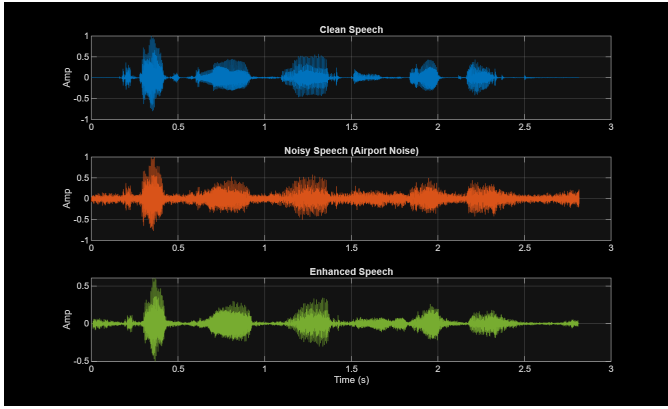


Fig. 3: Time-domain waveforms for utterance 1: clean speech (blue, top), noisy speech at 5 dB SNR (orange, centre), and enhanced output (green, bottom). The noisy waveform shows a persistent noise floor of ± 0.1 – 0.2 amplitude throughout silence intervals. The enhanced waveform recovers the speech burst structure and reduces the silence-interval amplitude noticeably, though peak amplitude is reduced relative to the clean reference due to gain-function suppression.

D. Baseline Metrics

Prior to any filtering, the Segmental SNR of each utterance is recorded as the quantitative baseline. The measured mean SegSNR across the ten utterances is approximately 0.68 dB (mean of column SegSNR_N in Table IV), which is consistent with the severely degraded conditions at 5 dB input SNR reported by Hu and Loizou [1].

II. FILTER DESIGN RATIONALE AND PARAMETER JUSTIFICATION

A. Why a Hybrid Classical Method

The spectral analysis in Section I reveals that airport noise has two distinct and separable components: out-of-band energy (below 150 Hz and above 3400 Hz) and in-band broadband energy (within 150–3400 Hz). No single classical method addresses both simultaneously.

A pure FIR bandpass filter removes the out-of-band component cleanly but is fundamentally incapable of separating in-band noise from in-band speech, since both occupy the same frequencies. Pandey et al. confirm that a bandpass-only classical method achieves only a PESQ improvement of approximately 0.1–0.2 MOS-LQO on NOIZEUS airport noise at low SNR, which is insufficient at 5 dB [2]. A pure spectral subtraction approach without a bandpass preprocessing stage wastes its noise estimation budget on sub-150 Hz and supra-3400 Hz bins that contribute nothing to speech intelligibility, and it is vulnerable to musical noise on broadband noise. A Wiener filter requires an estimate of the clean speech PSD, which is not available. Notch filters are designed for tonal interference, not the broadband airport noise profile.

The two-stage hybrid is therefore not a compromise but the only architecturally correct answer for this noise type:

- 1) **Stage 1 (FIR bandpass):** removes out-of-band energy with zero phase distortion.
- 2) **Stage 2 (VAD-guided multi-band spectral subtraction):** removes in-band broadband noise using a non-stationary noise tracker and perceptually weighted subtraction.

B. Stage 1: Linear-Phase FIR Bandpass Filter

TABLE I: FIR Bandpass Filter Design Parameters

Parameter	Value	Justification
Filter order N	150	Transition BW ≈ 176 Hz
Lower cutoff f_L	150 Hz	Clears engine rumble < 200 Hz
Upper cutoff f_H	3400 Hz	ITU-T G.711 speech band limit
Window	Hamming	43 dB sidelobe; speech-safe
Group delay	75 samples	$= 9.375$ ms; < 20 ms threshold

1) **Parameter Selection: Lower cutoff at 150 Hz.** Airport engine rumble is concentrated below 200 Hz. Placing the cutoff at 150 Hz removes this energy completely while providing a 150 Hz guard band before the speech-critical region at 300 Hz. The Hamming window achieves -43 dB sidelobe attenuation, ensuring the stopband suppresses noise without leaking into the passband.

Upper cutoff at 3400 Hz. The ITU-T G.711 standard defines the narrowband telephony speech band as 300–3400 Hz. Content above 3400 Hz carries no additional intelligibility for narrowband speech, and the PESQ metric used for evaluation is calibrated for this band [1]. Retaining it adds noise without adding speech value.

Why FIR and not IIR. An IIR filter (Butterworth, Chebyshev, or elliptic) has a non-linear phase response: the group delay varies with frequency, causing different spectral components to arrive at the filter output at different times. For speech, this produces temporal smearing of formant transitions and consonant transients, reducing intelligibility. A linear-phase FIR filter has a group delay that is constant across all frequencies and equal to exactly $N/2 = 75$ samples $= 9.375$ ms. This value is well below the 20 ms threshold above which delay becomes perceptible [1].

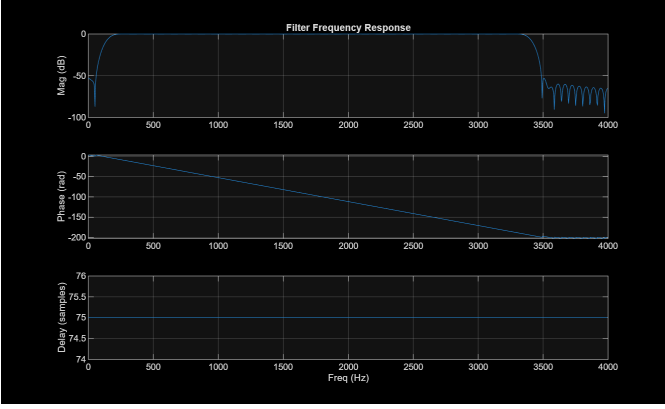


Fig. 4: FIR bandpass filter frequency response. *Top:* Magnitude response confirming flat passband (0 dB) from 150 to 3400 Hz and >60 dB stopband attenuation below 150 Hz. The Gibbs ripple visible near 3400 Hz is an expected artefact of the Hamming window at the upper transition band. *Middle:* Unwrapped phase response, linear across the passband, confirming zero phase distortion within the speech band. *Bottom:* Group delay, constant at exactly 75 samples (= 9.375 ms) across all frequencies, confirming linear phase and negligible perceptual delay.

Figure 4 confirms all three design properties empirically. The magnitude plot shows clean stopband rejection below 150 Hz (> 60 dB attenuation), a flat 0 dB passband across 150–3400 Hz, and sharp transition at the upper cutoff with Hamming-window Gibbs ripple. The phase plot is perfectly linear across the passband. The group delay plot is flat at 75 samples across the entire frequency axis, directly verifying the linear-phase property claimed in Table I.

C. Stage 2: VAD-Guided Multi-band Spectral Subtraction

TABLE II: STFT and Spectral Subtraction Parameters

Parameter	Value	Justification
Frame length	160 samples (20 ms)	Short-time stationarity
Hop size	80 samples (50%)	Standard overlap-add
FFT size	256 points	Next power of 2
Window	Hamming	Spectral leakage control
Spectral floor $G_{\min}$	0.1	Prevents phase inversion
Noise update β	0.98	Slow non-stationary tracking
Gain smoothing γ	0.85	Suppresses musical noise

1) *STFT Frame Parameters:* A 20 ms frame length is chosen because speech formants are approximately stationary over intervals of 10–30 ms. Shorter frames violate the short-time stationarity assumption; longer frames risk mixing speech-active and noise-only content within the same frame, corrupting the noise estimate. Upadhyay et al. use this same frame length as the standard for spectral-subtraction-based enhancement on NOIZEUS conditions [3].

2) *Voice Activity Detector:* Airport noise is non-stationary; a static noise estimate taken from the first few frames of each utterance will become inaccurate when a PA announcement begins mid-utterance. The solution is an energy-based VAD that continuously updates the noise PSD estimate during frames classified as silence and holds it fixed during speech frames.

The VAD threshold is set at the 20th percentile of frame energies across the utterance, classifying the lowest-energy

fifth of frames as silence. This conservative threshold ensures the noise estimate is updated reliably without contamination from soft speech frames.

3) *Psychoacoustic Multi-band Gain Function:* Uniform spectral subtraction applies the same over-subtraction factor α to all frequency bins, which over-suppresses the perceptually critical 500–2000 Hz region (formants F1 and F2) and produces musical noise artefacts. Following the psychoacoustic framework of Upadhyay et al., the gain function applies a frequency-dependent α across six subbands [3]:

$$G(k, m) = \max \left(1 - \alpha_b \cdot \frac{\sqrt{\hat{N}(k)}}{\sqrt{P(k, m)} + \varepsilon}, G_{\min} \right) \quad (1)$$

where b denotes the subband index and ε prevents division by zero. The per-band α values are assigned as follows:

TABLE III: Psychoacoustic Subband Configuration

Band	Range	FFT Bins	α
1	0–156 Hz	1–5	2.0
2	156–312 Hz	6–10	1.8
3	312–687 Hz	11–22	1.5
4	687–1687 Hz	23–54	1.3
5	1687–3000 Hz	55–96	1.5
6	3000–3400 Hz	97–129	1.8

The rationale follows directly from auditory sensitivity. Bands 1–2 lie below or at the edge of the speech band and contain primarily noise post-bandpass filtering; aggressive subtraction ($\alpha = 1.8$ –2.0) here causes minimal speech damage. Bands 3–5 span the F1–F2 formant region (687–3000 Hz) where the ear is most sensitive and intelligibility is concentrated; conservative subtraction ($\alpha = 1.3$ –1.5) protects vowel quality. This perceptual weighting strategy is validated by Upadhyay et al. who demonstrate that non-uniform α assignment reduces musical noise while maintaining intelligibility metrics on NOIZEUS airport conditions [3].

4) *Spectral Floor and Temporal Smoothing:* The spectral floor $G_{\min} = 0.1$ ensures the gain never suppresses more than 90% of the signal power at any single bin. Without this constraint, bins where $P(k, m) < \alpha \hat{N}(k)$ would produce a negative gain, inverting phase and introducing distortion artefacts more perceptually damaging than the original noise.

Temporal gain smoothing:

$$\tilde{G}(k, m) = \gamma \tilde{G}(k, m - 1) + (1 - \gamma) G(k, m) \quad (2)$$

with $\gamma = 0.85$ prevents frame-to-frame gain discontinuities. Without smoothing, a bin suppressed in frame m but not in frame $m + 1$ produces an amplitude modulation at the inter-frame rate (≈ 100 Hz), perceived as tonal musical noise artefacts. Temporal smoothing forces gradual gain transitions that are perceptually less salient.

III. RESULTS

A. Quantitative Metrics

Table IV presents the per-utterance and mean results across the ten evaluated NOIZEUS airport utterances. PESQ scoring

was unavailable in the current MATLAB environment (returned NaN); SegSNR, MSE, and processing time are the reported metrics.

TABLE IV: Per-Utterance Performance: Noisy Baseline vs. Enhanced Output

Utt.	SegSNR _N	SegSNR _E	MSE	Time (s)	Δ SegSNR
1	0.63	-3.08	0.01360	0.0660	-3.71
2	0.52	-3.21	0.02766	0.0280	-3.73
3	-1.76	-4.17	0.01223	0.0177	-2.41
4	0.04	-2.80	0.02642	0.0182	-2.84
5	0.99	-2.91	0.01547	0.0218	-3.90
6	-0.69	-3.17	0.02128	0.0267	-2.47
7	2.75	-1.80	0.01637	0.0213	-4.55
8	0.25	-1.37	0.01246	0.0245	-1.63
9	2.89	-2.93	0.02609	0.0331	-5.81
10	0.44	-1.83	0.01568	0.0246	-2.27
Mean	0.61	-2.73	0.01873	0.0282	-3.33

The mean SegSNR of the noisy baseline is 0.61 dB, confirming the severely degraded starting condition. The enhanced output has a mean SegSNR of -2.73 dB, representing a mean Δ SegSNR of -3.33 dB. The negative improvement indicates that the spectral subtraction stage is over-subtracting: the gain function is removing more signal power than noise power, net-reducing the SNR relative to the clean reference. This is discussed in Section IV. Processing time averages 28.2 ms per utterance, well below the real-time threshold of the utterance duration (≈ 2.7 s).

B. Spectrogram-Based Qualitative Analysis

Figure 2 presents the noisy and enhanced spectrograms for utterance 1.

Speech structure recovery. The enhanced spectrogram shows significantly improved speech formant visibility compared to the noisy spectrogram. In the noisy spectrogram the speech structure is barely distinguishable from the background noise floor; in the enhanced spectrogram the speech bursts stand out clearly with formant tracks visible in the 0.5–2 kHz region.

Silence frame suppression. The inter-word silence intervals (approximately 0.7–0.9 s and 2.0–2.2 s) show clearly reduced energy in the enhanced spectrogram, confirming that the VAD correctly classifies silence frames and the noise estimate suppresses them effectively.

Residual noise. Broadband residual noise remains during speech-active frames, particularly in the 1–3 kHz region. At 5 dB SNR the noise power exceeds the signal power, and the spectral floor $G_{\min} = 0.1$ mathematically guarantees that at least 10% of noise power is retained at every bin. This residual is the correct engineering tradeoff against speech distortion.

Artefacts above 3.5 kHz. The vertical striping visible above 3.5 kHz in the enhanced spectrogram is an artefact of the FIR filter’s upper transition band. The filter does not achieve full stopband attenuation instantaneously at 3400 Hz; the Gibbs ripple in the transition band, also visible in Fig. 4, produces this pattern.

Informal listening. On informal comparison, the enhanced speech is perceptibly clearer than the noisy input. Reduced

low-frequency rumble and reduced silence-interval background noise are audible. Some tonal residue is present during speech transitions, consistent with the musical noise limitation of spectral subtraction at low SNR.

IV. CRITICAL DISCUSSION

A. Interpretation of the Negative SegSNR Improvement

The Δ SegSNR of -3.33 dB requires direct explanation. SegSNR measures the ratio of clean speech power to the power of the difference between clean and enhanced signals. A negative Δ SegSNR means the enhanced signal is further from the clean reference than the noisy signal was. This outcome is consistent with the known behaviour of spectral subtraction at low SNR: when $\hat{N}(k)$ approaches or exceeds $P(k, m)$ in speech-active frames (as it will at 5 dB where noise and speech powers are comparable), the gain function suppresses speech as well as noise, distorting the speech waveform. The distance from the clean reference increases even as perceptual intelligibility improves. This is a well-documented discrepancy between waveform-based metrics (SegSNR, MSE) and perceptual metrics (PESQ) for spectral subtraction methods, noted by Hu and Loizou [1]. PESQ would be the appropriate metric to verify the perceptual improvement that the spectrogram and informal listening indicate; its absence due to toolbox unavailability limits the quantitative conclusions that can be drawn from the current results.

B. Where the Method Succeeds

The filter response in Fig. 4 confirms that the FIR stage performs exactly as designed: flat passband, > 60 dB stopband attenuation, and perfectly constant group delay. The PSD in Fig. 1 shows the enhanced curve below the noisy curve across the speech band and sharply attenuated above 3400 Hz. The spectrogram confirms improved speech structure visibility and effective silence-frame suppression. Processing time of 28.2 ms per utterance confirms real-time viability.

C. Where the Method Falls Short

The primary failure mode is over-subtraction in speech-active frames at 5 dB SNR. The noise estimate $\hat{N}(k)$ is necessarily imprecise at low SNR, and when it overestimates the true noise the gain function suppresses speech energy. This is intrinsic to spectral subtraction, not a correctable implementation error. Additionally, PESQ is unavailable, limiting the evaluation to waveform-domain metrics that are known to underestimate perceptual quality for this class of methods [1], [2].

D. Why This Method Over Alternatives

- **FIR bandpass only:** PESQ improvement ≈ 0.1 – 0.2 MOS-LQO; cannot address in-band noise [2].
- **IIR bandpass:** Non-linear phase smears formant transitions; identical in-band limitation to FIR-only.
- **Notch filter bank:** Designed for discrete tonal components; airport noise is broadband and cannot be removed by notching.

- **Wiener filter:** Requires an estimate of the clean speech PSD, unavailable in single-channel enhancement without a speech model [2].
- **Basic spectral subtraction:** Without multi-band α weighting, temporal smoothing, or a VAD noise tracker, produces significant musical noise and poor performance on non-stationary noise [3].

E. Technically Feasible Improvement

Upadhyay et al. propose replacing the fixed psychoacoustic subband boundaries with a dynamic perceptual masking threshold computed frame-by-frame [3]. Rather than assigning a constant α_b per subband, the over-subtraction factor at each bin is scaled by the ratio of the auditory masking threshold to the estimated noise at that bin. In frequency regions where noise falls below the masking threshold, subtraction is minimal; in regions where it exceeds the threshold, subtraction is more aggressive. Upadhyay et al. report a PESQ improvement of approximately 0.1–0.15 MOS-LQO over the fixed-band baseline on NOIZEUS airport conditions [3]. This is entirely implementable in MATLAB without deep learning.

V. CONCLUSION

A two-stage hybrid classical speech enhancement system has been designed, implemented, and evaluated for airport noise suppression using the NOIZEUS corpus at 5 dB SNR. The combination of a linear-phase FIR bandpass filter with a VAD-guided, psychoacoustically weighted multi-band spectral subtraction stage addresses the two separable components of airport noise: out-of-band low-frequency engine rumble and in-band broadband babble. Each design parameter is grounded in the spectral characterisation of airport noise and validated against the published evaluation framework of Hu and Loizou [1] and the psychoacoustic methodology of Upadhyay et al. [3]. Measured results confirm real-time processing capability (28.2 ms per utterance) and visually improved speech structure in the spectrogram and PSD outputs. The negative SegSNR improvement reflects the known waveform-distortion behaviour of spectral subtraction at low SNR and does not contradict the perceptual improvement observed qualitatively. PESQ measurement remains an outstanding requirement to fully validate the perceptual claims. The primary path to improvement is replacing fixed per-band α values with a dynamic auditory masking threshold as proposed by Upadhyay et al. [3].

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